Milestone 1 Report: NLP Foundation and Semantic Similarity

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Objective

The objective of this milestone was to understand and implement the fundamental concepts of Natural Language Processing (NLP) and semantic similarity. The tasks included text preprocessing, handling missing data, generating text embeddings using TF-IDF and Word2Vec, computing cosine similarity between prompts and answer options, and evaluating ranking performance using Mean Average Precision at 3 (MAP@3).

Dataset Overview

The dataset consists of multiple-choice question-answer pairs. Each record contains:

- Prompt (question text)
- Five answer options (A, B, C, D, E)
- Correct answer label

The training dataset was used for analysis, embedding generation, similarity computation, and evaluation.

Text Preprocessing

Text preprocessing was performed to convert raw text into a clean and consistent format suitable for machine learning.

The following preprocessing steps were applied:

1. Conversion of all text to lowercase.
2. Removal of punctuation characters.
3. Tokenization of text into individual words.
4. Removal of English stop words.
5. Handling missing values by replacing null entries with empty strings.

Example:

Original Text:
"What is Artificial Intelligence?"

Processed Text:
"artificial intelligence"

This preprocessing reduced noise and improved the quality of the generated embeddings.

Exploratory Analysis

Frequency Distribution of Answers

The distribution of correct answers (A, B, C, D, E) was analyzed to understand class imbalance within the dataset.

The frequency count of each answer option was computed using the training dataset. The most frequent and least frequent answer classes were identified and used later for the majority-class baseline evaluation.

Vocabulary Analysis

The prompt column was cleaned by converting text to lowercase and removing punctuation. The cleaned prompts were tokenized using whitespace splitting.

The total number of unique words across all prompts was calculated to determine the vocabulary size of the dataset.

TF-IDF Embedding Generation

Term Frequency-Inverse Document Frequency (TF-IDF) was used as the baseline text representation technique.

A TF-IDF vectorizer with English stop-word removal was trained using the combined text from prompts and answer options.

TF-IDF assigns higher importance to words that are frequent within a document but relatively rare across the dataset.

Advantages:

- Simple and computationally efficient.
- Interpretable feature representation.
- Effective baseline for text similarity tasks.

The resulting TF-IDF matrix transformed textual content into numerical vectors suitable for similarity calculations.

Word2Vec Embedding Generation

Word2Vec embeddings were generated using the Gensim library.

The model was trained on tokenized text extracted from prompts and answer options. Sentence-level embeddings were obtained by averaging the word vectors corresponding to the tokens present in each sentence.

Advantages:

- Captures semantic relationships between words.
- Produces dense vector representations.
- Can identify similarity between words with related meanings.

Example:

king − man + woman ≈ queen

Word2Vec provided a semantic representation that complements the frequency-based TF-IDF approach.

Cosine Similarity

Cosine similarity was used to measure the similarity between a prompt and each answer option.

The cosine similarity between two vectors is defined as:

Cosine Similarity = $(A \cdot B) / (||A|| \times ||B||)$

Interpretation:

- 1.0 → Identical vectors
- 0.0 → No similarity
- Negative values → Opposite directions

For each question:

1. The prompt embedding was generated.
2. Embeddings for options A–E were generated.
3. Cosine similarity was computed between the prompt and each option.
4. Options were ranked according to similarity scores.

The option with the highest similarity score was considered the predicted answer.

Majority Class Baseline

A baseline model was constructed using answer frequencies from the training set.

The three most frequent answer labels were identified and used as fixed predictions for every question.

Prediction format:

1st Prediction → Most frequent answer

2nd Prediction → Second most frequent answer

3rd Prediction → Third most frequent answer

This baseline provided a simple benchmark against which semantic similarity methods could be compared.

Evaluation Metric: MAP@3

Mean Average Precision at 3 (MAP@3) was used to evaluate ranking performance.

For each question:

- If the correct answer appeared at Rank 1:
  AP@3 = 1.0
- If the correct answer appeared at Rank 2:
  AP@3 = 0.5
- If the correct answer appeared at Rank 3:
  AP@3 = 0.3333
- If the correct answer did not appear in the top 3 predictions:
  AP@3 = 0

MAP@3 is calculated as the mean of AP@3 scores across all questions.

This metric evaluates both correctness and ranking quality.

TF-IDF Similarity Pipeline

A complete TF-IDF ranking pipeline was implemented.

Pipeline Steps:

1. Fit TF-IDF vectorizer on prompts and options.
2. Transform prompts and options into TF-IDF vectors.
3. Compute cosine similarity between prompt and each option.
4. Rank options according to similarity.
5. Select top three predictions.
6. Calculate AP@3 for each question.
7. Compute overall MAP@3 across the training dataset.

The resulting MAP@3 score served as the primary performance metric for the baseline semantic retrieval system.

Key Learnings

Through this milestone, the following concepts were understood and implemented:

- Text cleaning and preprocessing techniques.
- Handling missing textual data.
- Tokenization and stop-word removal.
- TF-IDF feature extraction.
- Word2Vec semantic embeddings.
- Cosine similarity computation.
- Ranking-based answer selection.
- MAP@3 evaluation methodology.

- Majority-class baseline construction.

These foundational NLP techniques form the basis for more advanced semantic retrieval and question-answering systems that will be explored in future milestones.

Conclusion

Milestone 1 successfully established the NLP foundation required for semantic similarity tasks. The implementation demonstrated how textual data can be transformed into numerical representations, how similarity between texts can be measured using cosine similarity, and how ranking performance can be evaluated using MAP@3.

The TF-IDF and Word2Vec approaches provided valuable insight into both frequency-based and semantic representations of text. These techniques will serve as the baseline for future improvements using advanced embedding models and retrieval methods.